

MAT 1013: Discrete Mathematics

Session 6: Natural Numbers, Divisibility, GCD, Primes and Factorization

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Topics

- Natural numbers and the well-ordering principle
- Divisibility and basic divisibility properties
- Even and odd integers
- Division algorithm
- Greatest common divisors and least common multiples
- Euclidean algorithm
- Prime numbers and composite numbers
- Fundamental Theorem of Arithmetic
- Applications and proof problems

Learning Outcomes

By the end of this handout, students should be able to:

- Describe and use the Well-Ordering Principle
- Use the language and notation of divisibility correctly.
- Apply the division algorithm to compute quotients and remainders.
- Compute greatest common divisors (gcd) using factor listing and the Euclidean algorithm.
- Compute least common multiples (lcm).
- Solve problems involving gcd and lcm.
- Identify prime and composite numbers and use prime factorization to solve problems.
- Write simple proofs involving divisibility, gcd and lcm.
- Apply divisibility ideas to practical and computational problems.

1. Natural Numbers and the Well-Ordering Principle

The set of natural numbers is usually written as $\mathbb{N} = \{1, 2, 3, \dots\}$.

Note: Some authors include 0 in $\mathbb{N}$. In this course, unless otherwise stated, we use the above convention.

Well-Ordering Principle

Every nonempty subset of $\mathbb{N}$ has a least element.

This principle seems simple, but it is one of the foundations of many proofs involving natural numbers.

Examples

1. The set $A = \{n \in \mathbb{N} : n > 10\}$ has least element 11.
2. The set $B = \{n \in \mathbb{N} : n^2 > 50\}$ has least element 8, since $7^2 = 49 < 50$ but $8^2 = 64 > 50$.

Activity 1: Find the Least Element

Find the least element of each set.

1. $A = \{n \in \mathbb{N} : 3n + 2 > 20\}$
2. $B = \{n \in \mathbb{N} : n^2 \geq 100\}$
3. $C = \{n \in \mathbb{N} : 5 \mid n \text{ and } n > 31\}$
4. $D = \{n \in \mathbb{N} : n! > 10^n\}$
5. $E = \{n \in \mathbb{N} : 4 \mid n \text{ and } 6 \mid n\}$
6. $F = \{n \in \mathbb{N} : n > 4\sqrt{n}\}$

Activity 2:

Find the smallest positive integer n such that $2n$ is a perfect square and $3n$ is a perfect cube, if such an integer exist.

I.e., Find the smallest element of the set $\{n \in \mathbb{N} : 2n = a^2 \text{ and } 3n = b^3 \text{ for some } a, b \in \mathbb{N}\}$, if it is not empty.

Activity 3: Caution!

Well-ordering principle is very special to $\mathbb{N}$. For example, it is not true in $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$.

Example: Show that the subset of positive real numbers, $\{x \in \mathbb{R} : x > 0\}$ has no least element.

Hint: Proof by contradiction.

Principle of Mathematical Induction

Principle of mathematical induction is a consequence of well-ordering principle.

Principle of Mathematical Induction

Let $P(n)$ be a statement depending on a natural number n . Suppose that:

1. $P(1)$ is true.
2. For every $k \in \mathbb{N}$, if $P(k)$ is true, then $P(k + 1)$ is true.

Then $P(n)$ is true for every $n \in \mathbb{N}$.

Proof. Suppose the two hypotheses of induction hold. Let

$$S = \{n \in \mathbb{N} : P(n) \text{ is false}\}.$$

We need to show that S is the empty set.

We prove the result using the Well-Ordering Principle.

Recall that the Well-Ordering Principle states that every non-empty subset of $\mathbb{N}$ has a least element.

Suppose, for contradiction, S is not empty.

Because, $S \neq \emptyset$, by the Well-Ordering Principle, S has a least element.

Let this least element be m .

Thus $P(m)$ is false, and m is the smallest natural number for which $P(m)$ is false.

Since $P(1)$ is true by the base step, we cannot have $m = 1$.

Therefore, $m > 1$.

Hence $m - 1 \in \mathbb{N}$.

Since m is the least element of S , no natural number smaller than m belongs to S .

In particular, $m - 1 \notin S$.

Therefore $P(m - 1)$ is true.

But by the inductive step, if $P(m - 1)$ is true, then $P(m)$ is true.

This contradicts the fact that $P(m)$ is false.

Therefore our assumption was wrong.

Hence S must be empty.

Thus there is no natural number n for which $P(n)$ is false.

Therefore $P(n)$ is true for every $n \in \mathbb{N}$. □

Activity 4: Strong Induction

In ordinary induction, to prove $P(k + 1)$, we assume only $P(k)$. Sometimes this is not enough. For example, to prove a statement about $k + 1$, we may need to use several earlier cases:

$$P(1), P(2), \dots, P(k).$$

This leads to a stronger-looking form of induction.

Strong Induction

Let $P(n)$ be a statement depending on a natural number n .

Suppose that:

1. $P(1)$ is true.
2. For every $k \in \mathbb{N}$, if $P(1), P(2), \dots, P(k)$ are all true, then $P(k + 1)$ is true.

Then $P(n)$ is true for every $n \in \mathbb{N}$.

Exercise: Prove the strong version of the induction principle.

2. Divisibility

Definition: Divisibility

Let $a, b \in \mathbb{Z}$, with $a \neq 0$. We say that a divides b , written $a \mid b$, if there exists an integer k such that $b = ak$.

If a does not divide b , we write $a \nmid b$.

Examples

1. $3 \mid 12$, because $12 = 3 \cdot 4$.
2. $5 \nmid 17$, because there is no integer k such that $17 = 5k$.
3. $-4 \mid 20$, because $20 = (-4)(-5)$.

Activity 5: Divisibility Check

Determine whether each statement is true or false.

- | | |
|------------------|------------------|
| 1. $7 \mid 84$ | 4. $13 \mid 156$ |
| 2. $11 \mid 121$ | 5. $6 \mid 0$ |
| 3. $9 \mid 100$ | 6. $0 \mid 6$ |

Basic Divisibility Properties

Let $a, b, c \in \mathbb{Z}$, with $a \neq 0$.

1. If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.
2. If $a \mid b$ and $a \mid c$, then $a \mid (b - c)$.
3. If $a \mid b$, then for every integer m , $a \mid mb$.
4. For $a, b \neq 0$, if $a \mid b$ and $b \mid c$, then $a \mid c$.

Worked Proof

Claim: For all $b, c \in \mathbb{Z}$ and for all $a \in \mathbb{Z}$ with $a \neq 0$, if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.

Proof. Let $a, b, c \in \mathbb{Z}$ with $a \neq 0$.

Suppose $a \mid b$ and $a \mid c$.

Since $a \mid b$, there exists $m \in \mathbb{Z}$ such that $b = am$.

Since $a \mid c$, there exists $n \in \mathbb{Z}$ such that $c = an$.

Therefore, $b + c = am + an = a(m + n)$.

Because, $m, n \in \mathbb{Z}$, $m + n \in \mathbb{Z}$.

It follows that $a \mid (b + c)$.

Thus, the claim. □

Activity 6:

Prove the basic properties of divisibility.

Activity 7: The Tiling Problem

A rectangular floor has dimensions $84 \text{ cm} \times 126 \text{ cm}$. We want to cover the floor using identical square tiles, with no cutting and no gaps.

1. What side lengths are possible for the square tiles?
2. What is the largest possible side length?
3. How many square tiles are needed if the largest possible square tile is used?

3. Even and Odd Integers

Definitions

Definition : Even and Odd integers

1. An integer n is called **even** if there exists an integer k such that $n = 2k$.
I.e., $2 \mid n$.
2. An integer n is called **odd** if there exists an integer k such that $n = 2k + 1$. I.e., $2 \nmid n$.

For any integer n , one and only one of the above holds.

Activity 8:

Verify that if $n = 2k + 1$, as in the definition, then n is not divisible by 2.

Challenge: Conversely, if n is not divisible by n , then there is an integer k such that $n = 2k + 1$.

Examples

1. $18 = 2(9)$, so 18 is even.

2. $17 = 2(8) + 1$, so 17 is odd.

Activity 9: Parity Table

Complete the table.

Expression	Result	Reason
even + even		
even + odd		
odd + odd		
even $\times$ even		
even $\times$ odd		
odd $\times$ odd		

Worked Proof

Claim. The sum of two odd integers is even.

Proof. Let m and n be odd integers.

Then there exist integers a and b such that $m = 2a + 1$ and $n = 2b + 1$.

Then $m + n = (2a + 1) + (2b + 1) = 2a + 2b + 2 = 2(a + b + 1)$.

Since $a, b \in \mathbb{Z}$ we have that $a + b + 1 = k \in \mathbb{Z}$.

Thus, $m + n = 2k$ with $k \in \mathbb{Z}$, and hence $m + n$ is even. □

Activity 10:

Prove all the claims in the “Parity Table”.

4. The Division Algorithm

Division Algorithm

Let $a \in \mathbb{Z}$ and $b \in \mathbb{N}$. Then there exist unique integers q and r such that $a = bq + r$, where $0 \leq r < b$.

Here q is called the quotient and r is called the remainder.

Examples

1. Parity (Even / Odd)

Every integer has remainder 0 or 1 when divided by 2. Therefore every integer is either even or odd.

$$n = 2q \text{ or } n = 2q + 1$$

2. $37 = 5(7) + 2$, so when 37 is divided by 5, the quotient is 7 and the remainder is 2.

3. $100 = 9(11) + 1$, so when 100 is divided by 9, the quotient is 11 and the remainder is 1.

Activity 11: Quotient and Remainder

For each, find integers q and r such that $a = bq + r$, $0 \leq r < b$.

1. $a = 45$, $b = 7$

3. $a = -2025$, $b = 13$

2. $a = 121$, $b = 11$

4. $a = -1000$, $b = 37$

Activity 12: Remainders Modulo 3

Every integer has one of the following forms: $n = 3q$, $n = 3q + 1$, $n = 3q + 2$.

Use this to show the following observations

1. Any integer n can be written in the form $3k$ or $3k \pm 1$, where $k \in \mathbb{Z}$.
2. If n is an integer that is not divisible by 3, then the remainder when n^2 is divided by 3 is always 1.

5. Greatest Common Divisor

Definition : Greatest Common Divisor (GCD)

Let $a, b \in \mathbb{Z}$, not both zero. A positive integer d is called the **greatest common divisor** of a and b , written $\gcd(a, b)$, if:

1. $d \mid a$ and $d \mid b$;
2. if $c \mid a$ and $c \mid b$, then $c \leq d$.

Example by Listing Factors

Find $\gcd(18, 24)$.

Positive divisors of 18: 1, 2, 3, 6, 9, 18.

Positive divisors of 24: 1, 2, 3, 4, 6, 8, 12, 24.

Common positive divisors: 1, 2, 3, 6.

Therefore: $\gcd(18, 24) = 6$.

Activity 13: GCD by Listing

Find:

1. $\gcd(12, 18)$
2. $\gcd(20, 35)$
3. $\gcd(36, 48)$
4. $\gcd(84, 126)$

Note : Listing divisors becomes inefficient when the numbers are large. We need a better method.

5. The Euclidean Algorithm

Key Idea

The Euclidean algorithm is based on the following principle:

$$\gcd(a, b) = \gcd(b, r), \text{ where } a = bq + r.$$

Why This Works

Suppose $a = bq + r$.

Then: $r = a - bq$.

Any common divisor of a and b also divides r . Similarly, any common divisor of b and r also divides a . Therefore a, b and b, r have the same common divisors, and hence the same greatest common divisor.

Worked Example 1

Find: $\gcd(126, 84)$.

Use division: $126 = 84(1) + 42$.

Then: $84 = 42(2) + 0$.

Therefore: $\gcd(126, 84) = 42$.

Worked Example 2

Find: $\gcd(252, 198)$.

$$252 = 198(1) + 54.$$

$$198 = 54(3) + 36.$$

$$54 = 36(1) + 18.$$

$$36 = 18(2) + 0.$$

Therefore: $\gcd(252, 198) = 18$.

Activity 14: Euclidean Algorithm Practice

Compute using the Euclidean algorithm:

1. $\gcd(420, 126)$
2. $\gcd(987, 610)$
3. $\gcd(1001, 143)$
4. $\gcd(2026, 154)$

Note : The Euclidean algorithm is one of the oldest known algorithms and is still fundamental in computer science, number theory, and cryptography.

Activity 15: GCD as an integer combination

Use the above computations to determine integers m and n such that $\gcd(a, b) = na + mb$.

1. $\gcd(420, 126)$
2. $\gcd(987, 610)$
3. $\gcd(1001, 143)$
4. $\gcd(2026, 154)$

Activity 16:

Prove the following:

Let $a, b \in \mathbb{Z}$, both non-zero. If $d = na + mb$ for some integers n, m , then $\gcd(a, b) \mid d$ and in particular $\gcd(a, b) \leq d$.

6. Relatively Prime Integers

Definition: Relatively Prime Integers

Two integers a and b are called **relatively prime** or **coprime** if $\gcd(a, b) = 1$.

Examples

1. $\gcd(8, 15) = 1$, so 8 and 15 are relatively prime.
2. $\gcd(12, 18) = 6$, so 12 and 18 are not relatively prime.

Activity 17: Consecutive Integers

Compute:

$$\gcd(5, 6), \quad \gcd(10, 11), \quad \gcd(99, 100).$$

What pattern do you observe?

Claim : For every integer n , $\gcd(n, n + 1) = 1$.

Proof. Let $d = \gcd(n, n + 1)$. Then $d \mid n$ and $d \mid (n + 1)$.

Therefore: $d \mid ((n + 1) - n)$.

Hence: $d \mid 1$.

Since d is positive, we must have $d = 1$.

Therefore: $\gcd(n, n + 1) = 1$. □

Activity 18:

Prove the following statements for integers $a, b \in \mathbb{Z}$.

1. $\gcd(a, b) = 1$ if and only if there are integers m and n such that $ma + nb = 1$.

7. Prime Numbers

Definition

An integer $p > 1$ is called **prime** if its only positive divisors are 1 and p .

Examples

2, 3, 5, 7, 11, 13 are prime.

4, 6, 8, 9, 10, 12 are composite.

Why is 1 Not Prime?

The number 1 has only one positive divisor: itself. It is also a special element of $\mathbb{Z}$; a *unit*. Therefore, 1 is not a prime. This convention is essential for uniqueness of prime factorization.

An integer greater than 1 that is not prime is called **composite**.

Activity 19:

If $n \in \mathbb{N}$ factors as $n = cd$, with $c, d \in \mathbb{N}$, then show that one of the factors must be less than or equal $\sqrt{n}$.

Deduce that if n is not a prime, then it has a prime factor less than or equal $\sqrt{n}$.

Testing for Primality

To test whether $n > 1$ is prime, it is enough to check divisibility by prime numbers less than or equal to $\sqrt{n}$.

Example

Is 97 prime?

Since $\sqrt{97} < 10$, we only check primes: 2, 3, 5, 7.

The number 97 is not divisible by any of these. Therefore 97 is prime.

Activity 18: Primes up to 100

Determine all the primes up to 100.

Activity 19: Primality Testing

Determine whether each number is prime.

- | | |
|--------|--------|
| 1. 101 | 3. 119 |
| 2. 191 | 4. 127 |

8. Fundamental Theorem of Arithmetic

Statement

Every integer greater than 1 can be written as a product of prime numbers, and this factorization is unique up to the order of the factors.

This is called the **Fundamental Theorem of Arithmetic**.

Examples

$$84 = 2^2 \cdot 3 \cdot 7.$$

$$126 = 2 \cdot 3^2 \cdot 7.$$

$$360 = 2^3 \cdot 3^2 \cdot 5.$$

Activity 20: Prime Factorization

Find the prime factorization of each number.

- | | |
|--------|---------|
| 1. 180 | 4. 540 |
| 2. 252 | 5. 2025 |
| 3. 360 | 6. 4096 |

9. Least Common Multiple

Definition

Let $a, b \in \mathbb{N}$. A positive integer m is called the **least common multiple** of a and b , written $\text{lcm}(a, b)$, if:

1. $a \mid m$ and $b \mid m$;
2. if $a \mid c$ and $b \mid c$, then $m \leq c$.

Using Prime Factorization

Example:

$$84 = 2^2 \cdot 3 \cdot 7, 126 = 2 \cdot 3^2 \cdot 7.$$

$$\text{Then: } \text{gcd}(84, 126) = 2 \cdot 3 \cdot 7 = 42, \text{ and } \text{lcm}(84, 126) = 2^2 \cdot 3^2 \cdot 7 = 252.$$

Relationship Between GCD and LCM

For positive integers a and b , $ab = \text{gcd}(a, b) \text{lcm}(a, b)$.

Activity 21: GCD and LCM

Find both $\text{gcd}(a, b)$ and $\text{lcm}(a, b)$.

- | | |
|-----------------------|----------------------|
| 1. $a = 181, b = 243$ | 3. $a = 84, b = 126$ |
| 2. $a = 45, b = 60$ | 4. $a = 72, b = 120$ |

Activity 22:

Show that for two positive integers a and b , if a and b are relatively prime, then $\text{lcm}(a, b) = ab$.

10. Exercises

1. (a) Determine whether $12 \mid 144$.
(b) Determine whether $15 \mid 2025$.
(c) Find all positive divisors of 60.
(d) If $a \mid b$, must $a \mid b^2$? Explain.
(e) If $a \mid b^2$, must $a \mid b$? Give a proof or a counterexample.
2. Find q, r such that $a = bq + r$, where $0 \leq r < b$.

(a) $a = 79, b = 6$	(c) $a = -250, b = 17$
(b) $a = 31, b = 32$	(d) $a = -1001, b = 29$
3. A rectangular garden measures $84 \text{ m} \times 126 \text{ m}$.
It is to be divided into identical square plots with each side being a positive integer in meters, with no leftover land.

(a) What is the largest possible side length of each square plot?

- (b) How many such square plots are needed?
 - (c) If the side length must be at least 10 m, what side lengths are possible?
4. Use the Euclidean algorithm to determine the gcd.
- (a) $\gcd(144, 96)$
 - (b) $\gcd(301, 471)$
 - (c) $\gcd(391, 299)$
 - (d) $\gcd(1729, 1001)$
5. Find the prime factorization.
- (a) 924
 - (b) 1155
 - (c) 1728
6. Write complete proofs.
- (a) Prove that if $a \mid b$ and $a \mid c$, then $a \mid (b - c)$.
 - (b) Prove that if $a \mid b$, then $a \mid bc$ for every integer c .
 - (c) Prove that the sum of two even integers is even.
 - (d) Prove that the product of two odd integers is odd.
 - (e) Prove that if n is odd, then n^2 is odd.
 - (f) Prove that for every integer n , $n(n + 1)$ is even.
 - (g) Prove that for every integer n , $\gcd(n, n + 1) = 1$.
 - (h) Prove that if n is divisible by 6, then n is divisible by 2 and 3.
 - (i) Prove that if $a \mid b$ and $b \mid a$, then $a = \pm b$.
 - (j) Prove that if p is prime and $p \mid a$, then $p \mid ab$.
7. Some challenging problems:
- (a) Prove that among any three consecutive integers, one is divisible by 3.
 - (b) Prove that for every integer n , $n^3 - n$ is divisible by 6.
 - (c) Prove that if n is odd, then $n^2 \equiv 1 \pmod{8}$.
(Find out what this means!)
 - (d) Find all integers n such that $n \mid n + 5$.
 - (e) Find all positive integers n such that $\gcd(n, 12) = 4$.
 - (f) Use the Euclidean algorithm to find $\gcd(6765, 4181)$. What do you notice about these numbers?
 - (g) Show that if a and b are relatively prime, then $\gcd(a + b, a) = 1$.
 - (h) Let $a, b \in \mathbb{N}$. Prove that $\gcd(a, b) \mid \text{lcm}(a, b)$.
8. Do a brief investigation on the following.
- (a) Why does the Euclidean algorithm work?
 - (b) Why the Euclidean algorithm terminates?
 - (c) What is the difference between a common divisor and the greatest common divisor?
 - (d) Why the Fundamental Theorem of Arithmetic requires 1 not to be prime?
 - (e) Why is prime factorization unique?
 - (f) How are gcd and lcm used in scheduling problems?
 - (g) How do prime numbers appear in cryptography?
 - (h) How do divisibility, gcd, primes, and factorization connect to each other?